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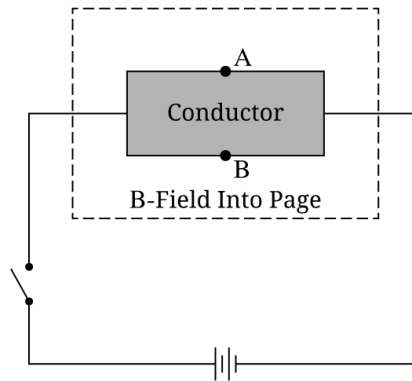
Brandon Furman 4/7

1. On what types of questions did you do well? What key content and/or skills were needed for success?

I did well on most of the problems. I struggled with problems that didn't have an immediate equation on the equation sheet to make it easier.

2. Reflect on 3 questions that you had difficulty with. For each question, do the following:

- Provide a screenshot of the question.
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- Explain what made this question/problem difficult for you.
 - What did you not understand or what made this question/problem challenging?
 - I had trouble with this problem because I didn't know the relationship between current density and magnetic field.
 - What can you remember or be sure to look out for next time?
 - I can remember to think of integrating around a surface to use Ampere's Law to find the enclosed current.
- Explain how you arrived at the correct answer, and show your work, as necessary.
 - By symmetry, we can integrate $\mu_0 \int \int J(x, y) dy dx$, which is equivalent to $\mu_0 I_{enc}$. We can do this double integral method to find that B becomes $\frac{B}{8}$.
 - What resources were helpful for you to answer the question/problem correctly?
 - Using the equation sheet, I could use the equation $\oint B \cdot dl = \mu_0 I_{enc}$
- Provide a screenshot of the question.



A rectangular metallic conductor is connected by copper wires in series with a battery and a switch as shown. The conductor lies within a region containing a uniform magnetic field directed into the page. A voltmeter is connected across points A and B such that if Point A is at a higher potential, the voltmeter will read a positive value. With the switch open, the voltmeter reads 0 V. A short time after the switch is closed what is the sign of the value the voltmeter reads?

☐ A A positive value

☐ B Zero

☒ C A negative value



☐ D A value alternating between positive and negative

- Explain what made this question/problem difficult for you.
 - What did you not understand or what made this question/problem challenging?
 - I had trouble with this problem because I didn't know what the significance of the magnetic field was if the conductor was stationary.
 - What can you remember or be sure to look out for next time?
 - I can remember that currents are the flow of electrons moving, which still get deflected by magnetic fields.
- Explain how you arrived at the correct answer, and show your work, as necessary.
 - By remembering that electrons get deflected by the opposite of the right hand rule in a magnetic field, I can see that the electrons move from right to left across the conductor and get deflected up. This causes the voltmeter to read a negative value.
 - What resources were helpful for you to answer the question/problem correctly?

- I used the fact that conventional current is opposite that of the movement of electrons and that increased electron density decreases voltage to come to my conclusion.
- Provide a screenshot of the question.

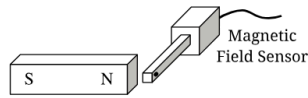


Figure 1

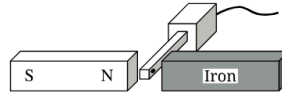


Figure 2

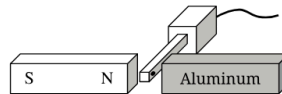


Figure 3

A student conducts an experiment using a sensitive magnetic field sensor. Three measurements are made with the sensor at the same fixed distance from the north pole of a stationary permanent bar magnet. In Trial 1 only the permanent magnet is near the sensor, as shown in Figure 1. In Trial 2 an initially unmagnetized iron bar is aligned with the bar magnet and positioned at an equal distance on the opposite side of the sensor, as shown in Figure 2. In Trial 3 an aluminum bar is aligned with the bar magnet and positioned at an equal distance on the opposite side of the sensor, as shown in Figure 3. The iron and aluminum bars have the same dimensions.

Let B_1 , B_2 , and B_3 be the magnitudes of the magnetic field strengths measured in trials 1, 2, and 3, respectively. Which of the following correctly compares B_1 , B_2 , and B_3 ?

(A) $B_1 < B_2 < B_3$

(B) $B_1 < B_3 < B_2$

(C) $B_3 < B_1 < B_2$

(D) $B_1 = B_3 < B_2$

✓

✗

- Explain what made this question/problem difficult for you.
 - What did you not understand or what made this question/problem challenging?
 - I didn't realize that paramagnetic materials will also create a weak magnetic field.
 - What can you remember or be sure to look out for next time?
 - I can remember that iron is ferromagnetic and aluminum is paramagnetic, and that ferromagnetism is akin to permanent magnetism.
- Explain how you arrived at the correct answer, and show your work, as necessary.
 - The permanent magnet creates a permanent magnet in the iron by aligning the magnetic domains. The permanent magnet causes the aluminum to create a weak magnetic field. Thus, $B_1 < B_3 < B_2$.
 - What resources were helpful for you to answer the question/problem correctly?

- To answer this question, it was helpful to try and sketch what the magnetic fields should look like.